

Editor-in-Chief

Dear Editor-in-Chief,

On behalf of my co-author and corresponding author, Professor Md. Jamal Uddin, I am pleased to resubmit our revised manuscript entitled:

"Association Between Internet Use and Unmet Need for Family Planning among Currently Pregnant Women in Bangladesh: Evidence from BDHS 2022"

for reconsideration for publication in your esteemed journal.

We sincerely thank you and the reviewers for your careful evaluation and constructive comments. We have thoroughly addressed all reviewer comments and substantially revised the manuscript accordingly. In particular, we revised the Results, Discussion, Policy Implications, and Conclusions; updated the descriptive and regression tables to improve clarity and consistency with standard scientific reporting; incorporated additional relevant references to strengthen the interpretation of our findings; and carefully corrected terminology, reporting, and methodological descriptions throughout the manuscript.

In response to the concerns regarding figure reproducibility, we would also like to clarify that **Figure 1 was not generated using artificial intelligence**. It was created using reproducible **LaTeX/TikZ** code, and the complete source code has been made publicly available in our GitHub repository to ensure transparency and reproducibility:

<https://github.com/muhammadsalek/BDHS-Unmet-Need-Analysis>

A detailed point-by-point response to all reviewer comments is provided in the accompanying **Response to Reviewers** document. For ease of evaluation, we have submitted the following files:

- Clean revised manuscript (final version)
- Revised manuscript with highlighted changes
- Response to Reviewers document

All revisions in the highlighted manuscript are clearly marked, and the corresponding page and line numbers have been provided in both the revised manuscript and the Response to Reviewers document to facilitate the review process.

We appreciate the time and effort devoted by you and the reviewers to evaluating our work. We hope that the revised manuscript satisfactorily addresses all concerns and will now be suitable for publication in your esteemed journal.

Sincerely,

Md. Salek Miah

On behalf of the Corresponding Author

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Reviewer 3

We sincerely thank the reviewer for the careful re-evaluation of our revised manuscript and for the constructive and encouraging comments. We appreciate the reviewer for identifying several inconsistencies between our previous response letter and the revised manuscript. Following your recommendations, we thoroughly rechecked the entire manuscript to ensure complete consistency between the response letter and the manuscript. All identified inconsistencies have now been corrected.

Comment 1

You told Reviewer 1 that you removed the C-statistic (AUC). However, Section 3.3 still discusses the ROC analysis, and Figure 3 is still a standalone ROC curve. Fix: Delete the ROC paragraph in 3.3 and completely remove Figure 3.

Response: [through manuscript]

We sincerely apologize for the inconsistencies between our previous response letter and the revised manuscript. We carefully rechecked the entire manuscript line by line and ensured that every promised revision has now been fully incorporated.

Thank you for identifying this inconsistency. We have now completely removed all references to the ROC curve and AUC (C-statistic) from the manuscript, including the corresponding paragraph in Section 3.3 and Figure 3, as recommended. Because this study was conducted within an explanatory (causal) framework rather than a predictive framework, we agree that reporting model discrimination statistics was unnecessary.

Comment 2

You told Reviewer 1 you deleted the non-essential global comparisons (like Sub-Saharan Africa). They are still in the second paragraph of your Introduction. Fix: Delete those sentences.

Response: [Page no:3, line no: 62-66]

Thank you for this valuable observation. We have revised the Introduction by removing the non-essential regional comparisons, including references to Sub-Saharan Africa and other region-specific prevalence estimates. The Introduction now follows a more focused structure, beginning

with the global burden of unmet need for family planning and then narrowing to the Bangladesh context, thereby providing a clearer and more relevant rationale for the present study.

Comment 3

You told the reviewer you removed temporal words like "today" to keep the timeline clear. The Introduction still says "...to over 60% today [8]". Fix: Change "today" to "as of 2022".

Response: [Page no: 3, line no:67-69]

Thank you. We have corrected this wording. The sentence now reads:

"Bangladesh has made substantial progress in contraceptive coverage over the past decades, with modern contraceptive use among married women increasing from less than 10% in the 1970s to over 60% by the early 2020s [8]."

Comment 4

You told Reviewer 2 you would clarify your study population. Yet, the very first sentence of your Discussion still says you studied "ever-married women" instead of "currently pregnant women". Fix: Change this immediately to prevent denominator confusion.

Response: [page no: 15, line no:336-337]

Thank you for identifying this inconsistency. We have revised the Discussion accordingly. The study population is now consistently described as **currently pregnant women** throughout the manuscript, including the Discussion, thereby avoiding any ambiguity regarding the analytical sample.

Comment 5

The Sample Size Mystery: Your text says your final sample size is 4,370, but the flowchart in Figure 1 ends at 4,448. Fix: Make sure the text matches the final box of your flowchart.

Response: [Page no: 26, Figure no: 1]

Thank you for pointing this out. The discrepancy resulted from a **typographical error** in the final box of Figure 1 rather than an inconsistency in the analytical dataset. We have corrected the flowchart, and the final analytical sample size is now consistently reported as **N = 4,370** throughout the manuscript.

Comment 6

Table 1 Discrepancy: In Table 1, the "No" and "Yes" rows for internet use add up to over 13,000 women, which is way larger than your actual study sample of 4,370. It looks like uncleaned data from the broader BDHS dataset accidentally stayed in those cells. Fix: Double-check and update those numbers.

Response: [Page no :11-12, line no: 284-287]

Thank you for carefully checking Table 1. We rechecked the frequencies and confirmed that Table 1 was generated from the final analytical sample (N = 4,370). The discrepancy was due to a **formatting/alignment error** in the previous version of the table rather than the use of an incorrect dataset. The table has been reformatted and carefully verified. The subgroup frequencies now correctly sum to the final analytical sample ($1,890 + 1,768 + 712 = 4,370$).

Overall Comment

Reviewer Strategy Comment: Right now, the science in the paper is solid, but the execution of authors revisions is high-risk... Once those text edits match your claims, this paper has a very high chance of acceptance.

Response: [through manuscript]

We sincerely appreciate the reviewer's encouraging assessment of the scientific quality of our study and the constructive recommendations. Following your advice, we performed a comprehensive line-by-line review of the entire manuscript to ensure complete consistency between the manuscript and the response letter. All inconsistencies identified by the reviewer have been corrected. We are grateful for your careful evaluation and valuable suggestions, which have substantially improved the clarity, consistency, and overall quality of the manuscript.

Reviewer 4

Reviewer Response

Comment: *Can these be shortened and written in a concrete way?*

Response: [Page no:4-5, line no:114-118]

Thank you for this helpful suggestion. We have revised the study objectives to make them shorter, more specific, and easier to follow while preserving the original aims. The objectives are now presented in a concise and parallel format.

Reviewer Response

Comment: *Paper?*

Response: [Page no: 5, line no: 119-120]

Thank you for this comment. We have revised the sentence to clearly state that the study provides evidence to inform policies and programs, rather than implying that the paper itself advances reproductive autonomy or service delivery. This revision better reflects the contribution of an observational study.

Comment 1

Not sure what this means. How?

Response: [Page no: 5, line no:119-120]

Thank you for this helpful comment. The sentence has been revised to more explicitly describe how the study contributes, using clearer and more precise language to avoid ambiguity.

Comment 2

Overall, the method section is too big which makes it hard to follow. There is no need to explain the entire design of the BDHS – this is one way to shorten it. In addition to this, the authors should consider other ways to make the design section shorter and readable. Also, Figure 2 is too hard to understand the message/s it wants to give.

Response: [page no: 5, line no: 123-129]

Thank you for these valuable suggestions. We have substantially revised the Methods section to improve its readability and conciseness. Specifically, we shortened the description of the BDHS survey design by retaining only the methodological details essential for understanding the study and removed repetitive methodological descriptions throughout the section.

Regarding Figure 2, we respectfully retained its original presentation because it is a theory- and literature-informed directed acyclic graph (DAG) developed *a priori* to represent the hypothesized relationships between the exposure, outcome, and potential confounders. This structure follows established causal inference principles and is consistent with previous studies using DAGs for confounder selection. We were concerned that modifying the structure or presenting it in an alternative format could alter the intended conceptual framework and causal assumptions underlying the study. Therefore, we retained the original DAG while revising the figure caption to better explain its purpose and interpretation.

Comment

This is the study design of BDHS. Please try to shorten it and write more about your own design – how you used the data and so on. Also, how important it is to give the table of the BDHS study design. Please consider.

Response: page no: [5, line no: 123-129]

Thank you for this valuable suggestion. We have substantially shortened the description of the BDHS survey methodology by retaining only the essential information on the study design and data source and referring readers to the official BDHS 2022 report for detailed survey methods. The Methods section now places greater emphasis on the design and analytical procedures of the

present study, including the study population, eligibility criteria, variable definitions, survey weighting, and statistical analyses, thereby improving the overall readability of the manuscript.

Reviewer Response

Comment: *How is this relevant to the objective/s of the paper?*

Response: [Page no: 7, line no: 178-180]

Thank you for this important comment. IPV was not considered a primary variable of interest but was included as a potential confounder based on the theoretical framework (DAG) and previous literature. Evidence suggests that IPV is associated with both women's internet use and reproductive decision-making, including unmet need for family planning. Therefore, adjusting for IPV helped reduce potential confounding in estimating the association between internet use and unmet need for family planning. We have clarified this rationale in the Methods section.

Comment 1

The paper explores the internet use and not the information source related to un/met need for FP. As the authors rightly pointed out here, the information sources are varied including internet. This needs to be clearly explained.

Response: [page no: 7, line no: 183-187]

Thank you for this important comment. The household media exposure variable was not intended to measure internet use or the source of family planning information. Instead, it was included as a household-level covariate representing access to traditional communication media (television, radio, newspapers, and mobile phones), which may facilitate exposure to general health information. We have revised the Methods section to clearly distinguish this variable from the main exposure, individual internet use.

Comment 2

So this is the source of the internet use for the mother? Need to explain this somewhere.

Response: [page no:7, line no: 185-187]

Thank you for this clarification request. No. Household media exposure was not used to determine the respondent's internet use. Individual internet use was measured independently using the BDHS question on whether the respondent had used the internet during the previous 12 months. In contrast, household media exposure represented access to traditional communication media (television, radio, newspapers, and mobile phones) and was included only as a household-level covariate. We have clarified this distinction in the Methods section.

Reviewer Comment

Should this be part of result section?

Response: [page no: 9-10, line no: 257-264, page no:27]

Thank you for your suggestion. We respectfully retained Figure 2 in the **Methods** section because it is a theory- and literature-informed directed acyclic graph (DAG) developed *a priori* to specify the hypothesized causal framework and guide confounder selection before statistical analysis. Since the DAG represents the study design rather than an analytical result, its placement in the Methods section is consistent with established causal inference practice. Therefore, we retained its position and clarified its purpose in the manuscript.

Reviewer Comment:

Is this heading appropriate for the texts/written beneath it? Please consider.

Response: [Through Result Section]

Thank you for this helpful suggestion. We carefully reviewed all Results section headings in relation to the corresponding text. Where appropriate, we revised the headings to ensure they more accurately reflect the analyses and findings presented under each subsection, thereby improving the clarity, consistency, and overall organization of the Results section.

Reviewer Comment

Is he not one of the authors? How come an author is acknowledged in a paper?

Response: [page no:19, line no:440-441]

Thank you for this important observation. We agree that authors should not be acknowledged in the Acknowledgements section. Accordingly, we have removed the acknowledgment to Dr. Md Jamal Uddin to comply with standard authorship and publication guidelines.

Reviewer Comment

Please complete the sentence. Also, it may be mentioned somewhere how the permission, if one is taken, was obtained. "Free cost of data" raise a question.

Response: [page no:19, line no:440-441]

Thank you for this helpful suggestion. We have revised the Acknowledgements section to clarify that the BDHS 2022 dataset was obtained after obtaining approval through the DHS Program data request system. We also replaced the phrase "free cost of data" with more appropriate wording.

Reviewer 5

Reviewer Comment – General Comment

This manuscript addresses an important and timely public health question by examining the association between internet use and unmet need for family planning among currently pregnant women in Bangladesh using nationally representative BDHS 2022 data. While the study has the potential to make a valuable contribution, it is not yet ready for publication, as several methodological and reporting issues need to be addressed. Four major concerns recur throughout the manuscript. First, the study population is described inconsistently, with currently pregnant women, ever-married women, and women of reproductive age used interchangeably across sections. Second, the Methods and Results are not fully aligned. For example, spatial analysis results are presented without a corresponding methodological description, and the reported stratified analyses do not clearly align with the stated study objectives. Third, some interpretations extend beyond the evidence presented. The manuscript uses causal language, includes ROC/AUC analysis despite not being a prediction study, and makes policy recommendations that are not directly supported by the study findings. Fourth, the reference list also requires careful revision, as it contains duplicate citations, verifiable bibliographic errors, and several references that do not appear relevant to the study topic. In addition, the manuscript requires substantial language editing. There are numerous grammatical errors and several instances of inappropriate terminology, which affect clarity and readability.

Response

Response: [Through manuscript]

We sincerely thank the reviewer for the careful, thorough, and constructive evaluation of our manuscript and for recognizing the importance and relevance of the research question. We have carefully considered all of the concerns raised and have substantially revised the manuscript accordingly.

Specifically, we have ensured consistent use of the study population throughout the manuscript by explicitly referring to **currently pregnant women** in the title, abstract, objectives, Methods, Results, Discussion, and Conclusions. The Methods and Results sections have been carefully aligned by expanding the methodological description of the spatial and stratified analyses and ensuring consistency with the stated study objectives. We also revised the manuscript to avoid causal language, removed the ROC/AUC analysis from the main manuscript (retaining it only as supplementary diagnostic material), and refined the interpretation and policy implications to

ensure they are fully supported by the observational findings. Furthermore, the reference list has been thoroughly reviewed to remove duplicate citations, correct bibliographic inaccuracies, and retain only references that are directly relevant to the study. Finally, the entire manuscript has undergone comprehensive language editing to improve grammar, clarity, consistency, and overall readability.

We believe that these extensive revisions have substantially strengthened the methodological rigor, reporting quality, and overall presentation of the manuscript, and we sincerely appreciate the reviewer's valuable suggestions.

Reviewer Comment 1

The sentence "...the internet users had higher odds of unmet need for family planning among all subgroups..." is internally inconsistent. It states that the association was observed among all subgroups, but only selected strata are listed, implying the association was not universal. Please clarify whether the association was consistent across all subgroups or limited to the specified ones. Consider reporting the adjusted odds ratios (aORs) with 95% confidence intervals for the most important subgroup findings. Also, correct the grammatical errors (e.g., "unmet need" instead of "unmet needs" and "those with media access" instead of "those who with media access").

Response [Page no: 2, line 42-49]

Thank you for this careful observation. We agree that the previous wording was inaccurate and could be interpreted as implying that the association was observed across all subgroups. We have revised the abstract to remove the phrase **"among all subgroups"** and now report only the key subgroups in which statistically significant associations were observed, together with the corresponding adjusted odds ratios (aORs) and 95% confidence intervals. We also corrected the grammatical errors, including replacing **"unmet needs"** with **"unmet need"**, and revised the wording to improve clarity, precision, and consistency.

Comment 2

The statement "Spatial findings reveal particular attention is required in Chittagong to improve equitable access to contraceptive services" is not supported by the Methods...

Response: [Page no: 2, line no:49-50]

Thank you for this valuable comment. We have revised the Results section to report only the descriptive spatial findings and removed the policy recommendation. In addition, we clarified the descriptive spatial analysis in the Methods section to ensure consistency between the Methods and Results.

Comment 3

The phrase "with substantial regional and socio-demographic heterogeneity" is not supported by the results presented.

Response: [page no: 2, line no: 53-56]

Thank you for this helpful suggestion. The conclusion has been revised to describe the observed regional and socio-demographic subgroup differences more accurately and conservatively.

Comment 4

The statement "These findings suggest that digital access alone may not reduce unmet need..." overinterprets the findings.

Response: [Page no: 2, line no: 53-56]

Thank you for this important comment. We agree that the previous statement extended beyond the evidence provided by this cross-sectional study. We have revised the conclusion to avoid causal interpretation and now limit our conclusions to the observed associations.

Comment 5

The recommendation to priorities interventions in "high-need eastern divisions, particularly Chittagong"...

Response: [Page no: 2, line no: 53-56]

Thank you for identifying this inconsistency. We have revised the abstract to remove this unsupported recommendation and now report only the observed descriptive spatial findings, ensuring consistency with the study results.

Comment 6

The statement "to reduce regional inequalities in contraceptive access" is not fully aligned with the study outcome.

Response: [Page no: 2, line no: 53-56]

Thank you for this valuable observation. Accordingly, we have revised the conclusion to align with the study outcome and removed statements referring to contraceptive access.

Comment 7

The recommendations would be stronger if they were more concise, specific, and directly linked to the study findings...

Response: [Page no: 2, line no: 53-56]

Thank you for this constructive suggestion. We have substantially revised the conclusion to make it more concise, directly supported by the study findings, and consistent with the observational nature of the analysis while maintaining its public health relevance.

Reviewer Comment 8

The keyword "Health services accessibility" is not well aligned with the study variables, as no direct measures of service accessibility (e.g., distance to a facility or travel time) were analyzed. Consider replacing it with keywords that better reflect the study exposure and outcome, such as Internet use, Family planning, Unmet need, Reproductive health, or Bangladesh.

Response: [page no: 3, line no:57]

Thank you for this valuable suggestion. We agree that **"Health services accessibility"** was not an appropriate keyword because the study did not directly assess healthcare accessibility. Accordingly, we have replaced this keyword with terms that more accurately reflect the study exposure, outcome, and context.

Comments on Introduction

Reviewer Comment 1

Paragraphs 1–3 contain considerable repetition... These overlapping statements should be consolidated to improve clarity and avoid redundancy.

Response: [page no: 3, line no: 62-69]

Thank you for this helpful observation. We agree that the Introduction contained repetitive descriptions of the global burden of unmet need for family planning and the progress in contraceptive use in Bangladesh. Accordingly, we have carefully revised and streamlined the Introduction by consolidating overlapping statements, removing redundant information, and retaining only one description of the contraceptive trend in Bangladesh with the most appropriate citation. These revisions have improved the clarity, conciseness, and overall flow of the Introduction.

Comment 2 (Paragraph 2)

The statement "Bangladesh has made notable improvement..." should be revised... The term "today" is imprecise...

Response: [page no: 3, line no: 67-69]

Thank you for this helpful suggestion. We agree that the term **"today"** was imprecise in a scientific manuscript. Accordingly, we have revised the sentence by replacing **"today"** with **"by the early 2020s"**, consistent with the cited reference, to improve clarity and temporal accuracy.

Additionally, the statement "BDHS 2022 shows 41.8% unmet need among married women of reproductive age" appears inconsistent with the published BDHS 2022 findings...

Response: [page no:5, line no: 131, page no: 17, line no: 399-399]

Thank you for this important observation. We agree that the estimate reported in the BDHS 2022 report refers to **all currently married women of reproductive age**, whereas our analysis was **restricted to currently pregnant women**. Therefore, the reported prevalence (41%) reflects the study's analytical sample rather than the national estimate reported by BDHS 2022. To avoid confusion, we have revised the manuscript to clearly specify that this estimate pertains only to **currently pregnant women** and have distinguished it from the national BDHS estimate reported for all currently married women. Additionally, we add this in limitation.

Reviewer Comment 3

Paragraph 4 should be made more concise and focused on the rationale of the study. Avoid broad background statements that are not directly relevant to the research question. In particular, the statement "Therefore, the digital expansion provides opportunities for women in Bangladesh" is too vague. Please specify what type of opportunities are relevant to this study (e.g., access to family planning information, reproductive health education, telehealth services, or health-seeking support) and provide appropriate supporting references.

Response [page no:4, line no:89-96]

Thank you for this valuable suggestion. We agree that the previous paragraph was broader than necessary and that the statement regarding digital expansion lacked specificity. Accordingly, we substantially revised Paragraph 4 to improve its focus and conciseness by emphasizing the study rationale. Specifically, we clarified that digital expansion may facilitate access to family planning information, reproductive health education, digital health resources, and health-seeking support, which are directly relevant to the research question. We also incorporated recent supporting literature to strengthen the rationale and improve the contextual background.

Comment 4 (Paragraphs 5 & 6)

The research gap should be stated more clearly and supported by existing evidence rather than asserted. Briefly summaries the limited evidence on internet use and family planning from Bangladesh or similar settings, then clearly explain what remains unknown. In particular, emphasise that the association between internet use and unmet need for family planning among currently pregnant women has not been adequately explored.

More importantly, the Introduction does not discuss currently pregnant women or the unmet need for family planning during pregnancy, despite these being the study population and primary

outcome. The background should be revised to better justify the choice of study population and outcome, ensuring that the rationale aligns with the study objectives.

Response: [Page no: 4, line no:96-105]

Thank you for this insightful and constructive comment. We agree that the research gap and study rationale required further clarification. Accordingly, we revised the Introduction by incorporating recent evidence from Bangladesh and similar settings on internet use and reproductive health to better support the existing knowledge base. We also explicitly clarified that, to our knowledge, the association between internet use and unmet need for family planning among **currently pregnant women** has not been adequately investigated using nationally representative data. In addition, we added a brief justification for focusing on currently pregnant women, explaining that, within the DHS revised framework, unmet need during pregnancy is determined based on the intendedness of the current pregnancy. These revisions improve the alignment between the background, study population, outcome, and study objectives.

Comment 5 (Objectives)

Objective 1: Clarify that the study population comprises currently pregnant women aged 15–49 years, rather than conflating all women of reproductive age with currently pregnant women.

Objectives 2 & 3: The objectives on inequalities and stratified analysis are not adequately supported by the Introduction or clearly described in the Methods. Briefly justify the relevance of geographic and socio-demographic inequalities in the Introduction and describe the corresponding analytical approach in the Methods. Also, rephrase Objective 3 as an aim (e.g., to examine whether the association varies across selected socio-demographic subgroups) rather than as a method.

Response [Page no: 4-5, line no:114-118]

Thank you for these valuable suggestions. We have revised the study objectives to clearly specify that the study population comprises **currently pregnant women** in Bangladesh. We also rephrased the third objective as a study aim rather than a methodological description. In addition, we strengthened the Introduction by briefly justifying the relevance of geographic and socio-demographic variations and clarified the corresponding stratified analytical approach in the Methods section. These revisions improve the alignment between the Introduction, study objectives, Methods, and Results.

Comment 6 (Language and terminology)

A substantial number of words appear to have been replaced with inappropriate synonyms, which obscures meaning (looks like AI-generated). Examples include: "deficient access", "prominence obstinate gaps", "Pregnancy intention evidence consents direct assessment", "subsidize to high unmet need", "plummeting unmet need", "remains limited explored", "digital assignation", "downgraded societies", and "archeologically have challenged inadequate contact". Please

undertake a full language edit, ideally by a fluent academic English editor, before resubmission. In several places, the intended meaning cannot be reconstructed with confidence.

Response: [through manuscript]

Thank you for this important observation. We sincerely apologize for the unclear wording in several parts of the manuscript. We have carefully revised the entire manuscript to improve the language, grammar, terminology, and readability. Inappropriate word choices and ambiguous expressions have been corrected throughout, and the manuscript has undergone a comprehensive academic English language edit to ensure clarity, precision, and consistency. We believe these revisions have substantially improved the overall quality and readability of the manuscript.

Comments on Methods

Comment 1 (Study Design and Data Source)

The entire "Study Design and Data Source" section requires thorough revision. Several methodological descriptions (e.g., sampling design, sampling frame, sample size, response rates, and survey procedures) are inconsistent with the official BDHS 2022 Final Report. Please carefully cross-check all methodological information against the BDHS 2022 report and revise accordingly.

Response: [page no: 5, line no: 123-129]

Thank you for this important comment. We carefully cross-checked the methodological descriptions against the official **BDHS 2022 Final Report** and revised the *Study Design and Data Source* section accordingly. To improve clarity and avoid unnecessary detail, we substantially shortened this section and retained only the information essential to describe the data source, while referring readers to the BDHS 2022 report for detailed survey design and sampling procedures. We also clarified the study population and analytical sample separately in the *Study Population and Eligibility Criteria* section, where we explicitly state that the analysis was restricted to **currently pregnant women aged 15–49 years** and describe the derivation of the final analytical sample. These revisions improve the accuracy, consistency, and readability of the Methods section.

Comment 2

The participant flow diagram (Figure 1) is inconsistent with the data source and study population described. The flowchart does not clearly illustrate the sample selection process from the BDHS 2022 dataset to the final analytical sample, and its layout appears AI-generated rather than following a standard epidemiological reporting format.

Response: [page no: 26, line no: 647-648]

Thank you for this valuable comment. We carefully reviewed and revised **Figure 1** to ensure that it clearly illustrates the participant selection process from the BDHS 2022 dataset to the final analytical sample of currently pregnant women. The flowchart has been updated to improve clarity and consistency with the study population described in the Methods section. We would also like to respectfully clarify that the flowchart was **not AI-generated**; it was manually created by the authors using **LaTeX (TikZ)**, and the complete source code has been made publicly available in our GitHub repository to ensure transparency and reproducibility. We hope these revisions and clarifications adequately address the reviewer's concern.

Comment 3

The manuscript requires comprehensive language editing. There are numerous grammatical errors, awkward sentence constructions, and non-standard academic expressions throughout the Methods section. Please revise the text thoroughly to improve clarity, readability, and the quality of the scientific writing.

Response:[through methods]

Thank you for this valuable suggestion. We carefully revised the entire Methods section to improve the language, grammar, sentence structure, and overall readability. Ambiguous and non-standard academic expressions were corrected throughout, and the text was edited to ensure greater clarity, consistency, and scientific precision. We believe these revisions have substantially improved the quality of the manuscript.

Comment 4

Please use terminology consistent with the BDHS 2022 report and standard DHS reporting. For example, replace terms such as ".....fecundity preferences" with the more commonly used "fertility preferences" (or another appropriate general term used in the BDHS report). Additionally, phrases such as ".....through 2017–2018 cartographical procedures" could not be identified in the BDHS 2022 report and should be verified or replaced with the exact terminology used in the official survey documentation.

Response: [page no: 5, line no:123-129]

Thank you for this valuable comment. We carefully reviewed the Methods section against the official **BDHS 2022 Final Report** and revised the terminology to ensure consistency with standard DHS reporting. The previously identified non-standard or inaccurate expressions have been removed or replaced with terminology consistent with the BDHS 2022 documentation. In addition, we substantially revised and simplified the *Study Design and Data Source* section to improve accuracy, clarity, and consistency with the official survey report. We believe these revisions have addressed the reviewer's concerns.

Comment 5

Please remove the sentence, "This binary classification captures recent digital access and reflects potential exposure to online health information and digital communication platforms." The Methods should focus only on how the variable was defined and measured.

Response: [Page no: 6, line no: 151-154]

Thank you for this helpful suggestion. We agree that the Methods section should focus solely on the definition and measurement of the exposure variable. Accordingly, we have removed the interpretative sentence and retained only the description of how internet use was defined and classified in the analysis.

Comment 6 (Survey Design and Weighting)

The statement *"This study followed the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines [37]"* is repetitive (Lines 159–162).

Response: [Page no:7-8, line no:201-204]

Thank you for this observation. We agree that this statement was repetitive. Accordingly, we have removed the duplicate statement and retained a single reference to the STROBE guidelines at the appropriate location in the manuscript to improve conciseness and avoid redundancy.

Reviewer Response

Comment 7

The statement "Stratified analyses were conducted by place of residence and maternal age to explore potential effect modification" appears inconsistent with the stated study objectives, particularly Objective 3. Please ensure that the analytical methods are fully aligned with the study objectives, or revise the objectives and analysis plan accordingly.

Response: [page no: 9, line no:239-241]

Thank you for this helpful comment. We agree that the previous description did not fully align with the revised study objectives. Accordingly, we updated both the study objectives and the Statistical Analysis section to ensure consistency. The Methods now clearly state that stratified analyses were performed by place of residence, administrative division, maternal age, age at first birth, maternal education, and household media exposure to examine whether the association between internet use and unmet need for family planning varied across selected socio-demographic subgroups. These revisions improve the alignment between the Objectives, Methods, and Results.

Reviewer Response

Comment 8

The Methods section states that spatial analysis was performed using R Studio (Version 4.5.1); however, no description of the spatial analytical methods is provided. Please include a detailed description of the spatial analysis.

Response: [Page no: 9, line no:242-244]

Thank you for this helpful suggestion. We agree that the spatial analytical approach required further clarification. Accordingly, we have expanded the Methods section to describe the descriptive spatial analysis performed in this study. Specifically, we clarified that division-level weighted prevalence estimates of unmet need for family planning were visualized using choropleth maps in R Studio (Version 4.5.1) with administrative boundary shapefiles. We also explicitly stated that the analysis was descriptive only and that no spatial autocorrelation or spatial regression analyses were performed. These revisions improve the transparency and reproducibility of the analytical methods.

Response

Comment 9

The c-statistic (AUC) appears unnecessary for the current analysis. As this is an explanatory study aimed at estimating associations rather than developing a predictive model, measures such as adjusted odds ratios with 95% confidence intervals are sufficient to address the research objectives. I recommend either removing this measure or providing a clear rationale for its inclusion and explaining how it contributes to addressing the study objectives.

Response: [page no: 9, line no:246-256]

Thank you for this valuable suggestion. We agree that the c-statistic (AUC) is not essential for an explanatory study focused on estimating associations rather than predictive performance. Accordingly, we have removed the ROC analysis, c-statistic (AUC), and all related descriptions and figures from the manuscript. The *Model Diagnostics and Assumption Checking* section has been revised to focus on diagnostic procedures directly relevant to logistic regression, including model calibration using the Hosmer–Lemeshow goodness-of-fit test and assessment of multicollinearity using variance inflation factors (VIFs). These revisions improve the methodological consistency of the manuscript and better align the analyses with the study objectives.

Comment 10

Please move the "Ethics Approval and Consent to Participate" section to follow the "Adjusted Confounders in Final Models" section to maintain the standard organisation of the Methods section.

Response [page no:9-10, line no: 257-269]

Thank you for this helpful suggestion. We have revised the organization of the Methods section accordingly. Specifically, the "**Ethics Approval and Consent to Participate**" section has been moved to follow the "**Adjusted Confounders in Final Models**" section to improve the logical flow and maintain the standard structure of the Methods section.

Comments on Results

Response

Comment 1 (Inconsistency between text and reported percentages)

The statement, "Overall, 41% of women revealed unmet need for family planning [Figure 3]," appears inconsistent with the study population. Please verify whether the prevalence is 41% or 0.41% and revise accordingly. Also, for whom?

Response: [Page no: 10, line no:272]

Thank you for this helpful comment. We have revised the sentence to clearly specify the study population and the reported prevalence. The prevalence is **41.0%**, not **0.41%**, and it refers to the **currently pregnant women** included in the analytical sample (n = 4,370). The Results section has been updated accordingly to avoid ambiguity.

Comment 2

The descriptive results require careful revision, as several statements do not align with the percentages reported in Table 1. For example, the manuscript states that women with no education had a higher unmet need (3.4%) than those with above secondary education (28.4%), whereas the reported percentages indicate the opposite. Similarly, the statement that female-headed households had a higher unmet need (14.99% vs. 26.27% in male-headed households) contradicts the reported values. Comparable inconsistencies are also evident for other variables, including media access. The authors should carefully verify all percentages against Table 1 and ensure that the narrative accurately reflects the direction and magnitude of the observed associations.

Response [page no: 10-12, line no:284-287]

Thank you for this careful observation. We have carefully reviewed all descriptive results against Table 1 and revised the text to ensure that the narrative accurately reflects the reported data and avoids misinterpretation. Ambiguous wording has been corrected throughout, and all reported percentages and their corresponding descriptions have been verified for consistency with the revised table.

Comment 3 (Parity)

Reviewer Comment:

The sentence, "Women who have higher parity (≥ 3 children) showed higher (11.7%) unmet need compared to women with no children (3.96%)," could be written more clearly. For example: "Women with three or more children had a higher prevalence of unmet need than women with no children (11.7% vs. 4.0%, $p < 0.001$).

Response: page no: 10, line no:275-282]

Thank you for this helpful suggestion. We have revised the sentence for improved clarity and readability. Following the revision of Table 1 to present row percentages, the Results section now reports the prevalence of unmet need using within-category percentages and describes the findings in a clearer and more consistent manner.

Comment 4 (Terminology)

Reviewer Comment:

Please replace "associated to" with "associated with" throughout the manuscript, as this is the appropriate terminology in epidemiological and scientific writing.

Response: [through manuscript]

Thank you for pointing this out. We have replaced "associated to" with "associated with" throughout the manuscript to ensure consistency with standard epidemiological and scientific writing conventions.

Reviewer Comment 5

Comment 5 (Table 2)

The organization of Table 2 is unclear and does not follow a standard scientific reporting format. The table should be comprehensively revised to improve its structure, readability, and presentation. One methodological question: Are the rural and urban models adjusted for the same covariates as the national model? If so, this should be explicitly stated in the table footnote. If not, the specific adjustment set for each model should be provided.

Response: [Page no: 13, line no:301-309]

Thank you for this valuable suggestion. We have comprehensively revised **Table 2** to improve its structure, readability, and consistency with standard scientific reporting. Specifically, we revised the table title, reorganized the column headings, and clearly separated the crude and adjusted models, including the national and stratified (rural and urban) analyses. We also expanded the table footnote to explicitly describe the adjustment strategy. The national model was adjusted for all predefined confounders, including place of residence, whereas the rural and urban stratified models were adjusted for the same covariates except place of residence, which was not applicable after stratification. These revisions improve the transparency and interpretability of the findings.

Reviewer Comment 6

Comment 6 (Table 3)

The title of Table 3 is incomplete and lacks clarity. Please revise and verify the title.

Response: [page no:13-14, line no:311-317]

Thank you for this helpful suggestion. We have revised the title of Table 3 to improve its clarity, completeness, and scientific accuracy. The updated title now clearly describes the purpose of the analysis and specifies that the results represent adjusted associations from stratified analyses across selected maternal and household characteristics.

Reviewer Comment 7 (ROC Analysis)

The inclusion of the ROC analysis ($AUC = 0.7783$) is not adequately justified. As the primary objective of this study is to examine associations rather than to develop or validate a prediction model, reporting the AUC adds limited value. Unless the authors explicitly justify its relevance to the study objectives, this analysis should be removed, or its purpose should be clearly explained.

Response: [through manuscript]

Thank you for this insightful comment. We agree that the primary objective of this study is to examine the association between internet use and unmet need for family planning rather than to develop or validate a prediction model. Following your suggestion, we have removed the ROC curve analysis and the corresponding AUC results from the manuscript. This revision improves the methodological consistency of the study and ensures that the presented analyses remain aligned with the study objectives.

Reviewer Comment 8

Comment:

The subheading "Spatial Inequality of Unmet Family Need" is unclear. Please revise the subheading and clearly describe the spatial analytical approach in the Methods section.

Response: [page no: 14, line no: 330]

Thank you for this valuable suggestion. We have revised the subsection title to improve clarity and scientific accuracy. The new subheading is **"Spatial Distribution of Unmet Need for Family Planning."** In addition, we have expanded the Methods section to clearly describe the spatial analytical approach, including the preparation of division-level prevalence estimates and the visualization of regional variation using thematic mapping. This revision improves methodological transparency and aligns the Results with the described analytical procedures.

Reviewer Comment 9

Comment:

The statement referring to Khulna and Barisal as western divisions is inaccurate and should be corrected.

Response: [page no: 15, line no: 331-332]

Thank you for identifying this geographical inaccuracy. We have corrected the description in the Results section. The revised text accurately reports the regional distribution of unmet need for family planning without incorrectly classifying Khulna and Barishal as western divisions.

Reviewer Comment 10 (Interpretation in Results)

Comment:

The statement, "...with programmatic attention needed in high-need eastern divisions" [Figure 4], is interpretative and should not be included in the Results section. This interpretation is more appropriate for the Discussion, Conclusion, or Recommendations section.

Response: [page no: 15, line no: 331-332]

Thank you for this important observation. Accordingly, we have removed the interpretative sentence from the Results section. The Results now present only the observed spatial distribution of unmet need, while the implications for programmatic attention have been reserved for the Discussion.

Comments on Discussion

Reviewer Comment

General Major Comment:

The Discussion requires substantial revision to improve scientific clarity, language, and interpretation. Several statements are difficult to follow due to grammatical issues, and some interpretations go beyond what the cross-sectional design can support. The authors should focus on interpreting the findings in light of the existing literature while avoiding speculative or causal language.

Response: [through discussion]

Thank you for this valuable and constructive comment. We have comprehensively revised the Discussion section to improve its scientific clarity, logical flow, and overall readability. The language has been carefully edited to enhance grammatical accuracy and precision throughout.

In addition, we have revised several statements to ensure that the interpretation of our findings is fully consistent with the cross-sectional nature of the study. Any wording that could imply causality or overstate the findings has been removed or replaced with more appropriate observational terminology (e.g., "associated with" rather than causal expressions). We have also strengthened the Discussion by interpreting our findings in the context of relevant published literature,

highlighting similarities and differences with previous studies while avoiding speculative conclusions. These revisions improve the scientific rigor and interpretability of the manuscript.

Reviewer Comment 1 (Paragraph 1)

The sentence, "This study examined the association between internet use and unmet need for family planning among ever-married women in Bangladesh," is inconsistent with the stated study population. Please clarify whether the analysis was conducted among ever-married women or currently pregnant women, and ensure that the study population is described consistently throughout the manuscript.

Response: [page no: 15, line no:336-340]

Thank you for identifying this inconsistency. We have carefully revised the manuscript to ensure that the study population is described consistently throughout. The previous wording referring to "ever-married women" has been corrected to "currently pregnant women," which accurately reflects the analytical sample used in this study. We have reviewed the entire manuscript and updated all relevant sections to maintain consistency in the description of the study population.

Reviewer Comment 2 (Paragraph 1)

The opening paragraph should provide a concise summary of the principal findings without repeating detailed numerical results. In addition, the statement that the model demonstrated acceptable discriminatory performance ($AUC = 0.78$) is not directly relevant to the study objectives and should be removed or clearly justified, as this is an association study rather than a prediction study.

Response: [page no: 15, line no:336-340]

Thank you for this constructive suggestion. We have revised the opening paragraph of the Discussion to provide a concise summary of the principal findings without repeating detailed numerical results presented in the Results section. In addition, following the reviewer's recommendation, we have removed the statement regarding the model's discriminatory performance (AUC), as the primary objective of this study was to examine associations rather than to develop or evaluate a prediction model. These revisions improve the clarity and focus of the Discussion while ensuring better alignment with the study objectives.

Reviewer Comment 3 (Interpretation of findings)

Several interpretations are speculative and are not directly supported by the study findings (e.g., internet use revealing latent demand, partner disagreement, digital inequality, or exposure to unverified online information). These interpretations should either be supported with stronger evidence from the literature or presented more cautiously as possible explanations rather than conclusions.

Response: [page no: 15-16, line no: 341-348, 354-365, line no: 374-384]

Thank you for this insightful comment. Accordingly, we have revised the relevant paragraphs to use more cautious language and to present these mechanisms as possible explanations rather than definitive conclusions. We have also strengthened the discussion by referring to relevant published literature where appropriate and have avoided causal or speculative interpretations that are not directly supported by our findings.

Response to Reviewer

Comment 5 (Strengths and Limitations)

The strengths section should be revised to include only genuine methodological strengths. Statements such as "forward stepwise conceptual and statistical methods" and "ROC offers objective evaluation of model insight" are not strengths of the study and require justification. Given the study's objectives, including ROC/AUC as a strength is not appropriate.

Response: [page no: 16, line no: 389-393]

Thank you for this constructive suggestion. We have revised the Strengths and Limitations section accordingly. Specifically, we removed the statements regarding the forward stepwise model selection approach and the ROC/AUC analysis, as these were not essential methodological strengths for the objectives of this association study. The revised section now focuses on the study's genuine strengths, including the use of nationally representative data, a large sample size, adjustment for relevant confounding variables, and stratified analyses that provide additional insights into rural–urban differences.

Reviewer Comment 6 (Limitations)

The limitations are generally appropriate; however, they should acknowledge the potential for residual confounding and discuss the possibility of overadjustment, particularly if contraception status was included in the adjusted models. In addition, as the study is based on self-reported (subjective) data, the authors should acknowledge the potential for reporting and recall bias, which may have influenced the findings.

Response to Reviewer: [page no: 17, line no:398-405]

Thank you for this valuable suggestion. We have revised the Limitations section accordingly. Specifically, we have acknowledged the possibility of **residual confounding** arising from unmeasured or unavailable variables in the DHS dataset. We have also noted that, although relevant confounding variables were included in the adjusted analyses, the possibility of **overadjustment** cannot be completely excluded, particularly for variables that may lie on the causal pathway. Furthermore, we have expanded the limitations to clarify that the study relied on **self-reported DHS data**, which may be subject to **reporting and recall bias**, potentially influencing the observed associations. These additions provide a more balanced and transparent discussion of the study limitations.

Response to Reviewer

Comment 7 (Policy and Programmatic Implications)

The policy implications should be more closely aligned with the study findings. Recommendations such as telehealth expansion, digital referral systems, digital literacy programs, and male engagement strategies extend beyond the evidence generated by this analysis. The policy recommendations should be presented more cautiously and explicitly linked to the study findings. Some recommendations appear overstated, given the evidence from this cross-sectional study, and should be appropriately tempered.

Response to Reviewer: [page no: 17,line no: 407-412]

Thank you for this thoughtful comment. We have substantially revised the **Policy and Programmatic Implications** section to ensure that the recommendations are directly aligned with the findings of our study. Specifically, we removed recommendations regarding telehealth expansion, digital referral systems, digital literacy programs, and male engagement strategies because these were not directly evaluated in the present analysis. We also revised the language to avoid causal or overstated interpretations and emphasized that the observed associations should be interpreted cautiously given the cross-sectional design. The revised section now focuses on evidence-based implications, including improving access to reliable family planning information through digital platforms, strengthening existing family planning services, acknowledging rural–urban and regional differences, and highlighting the need for further longitudinal and implementation

Response to Reviewer

Comment 8 (Conclusions)

The phrase, "...among ever-married women in Bangladesh," is inconsistent with the stated study population. Please clearly specify the study population and ensure it is described consistently throughout the manuscript.

The statement referring to "...high-quality contraceptive services" is not adequately supported by the study findings. Please provide the basis for this statement or revise it accordingly.

The conclusion should avoid policy recommendations that are not directly supported by the study findings. For example, recommending targeted programs in Chattogram (AOR: 1.98) is not fully justified, particularly when other divisions (e.g., Barishal (AOR: 3.24), Rajshahi (AOR: 3.30), and Sylhet (AOR: 2.98)) also demonstrated strong associations. The conclusion should instead focus on summarizing the principal findings and their public health implications without introducing unsupported recommendations. Several statements appear overstated and should be interpreted more cautiously.

Response to Reviewer: [page no: 17-18, line no: 413-420]

Thank you for these constructive comments. We have revised the **Conclusion** to ensure consistency with the study population by replacing "ever-married women" with "**currently pregnant women**" throughout the manuscript where applicable. We also removed statements regarding **high-quality contraceptive services** and **targeted programs in specific divisions**, as these recommendations were not directly supported by the findings of this cross-sectional analysis. Furthermore, the conclusion has been rewritten to focus on the principal findings, acknowledge the observational nature of the study, avoid causal or overstated interpretations, and emphasize the need for future longitudinal research to further investigate the observed associations.

Comment 1 (Reference relevance)

Response:

Thank you for this valuable observation. We have carefully reviewed the entire reference list and removed or replaced references that were not directly relevant to the study topic. Specifically, the cited references, including References 1, 41, 44, and 47, have been corrected and replaced with appropriate literature supporting the study objectives and corresponding statements.

Comment 2 (Irrelevant literature and incorrect citation)

Response:

Thank you for this important comment. We thoroughly reviewed all cited literature and removed or replaced references that were not directly relevant to the present study. References related to digital mental health and unrelated topics were substituted with appropriate family planning and reproductive health literature where necessary. In addition, the citation corresponding to Reference 48 was verified and replaced with a relevant source supporting the corresponding statement.

Comment 3 (Duplicate references)

Response:

Thank you for identifying these duplicate references. We have carefully reviewed the reference list, merged all duplicated entries, and updated the corresponding in-text citation numbering consistently throughout the manuscript.

Comment 4 (Reference accuracy)

Response:

Thank you for this helpful suggestion. We have carefully verified and corrected all references to ensure the accuracy and completeness of the bibliographic information, including authors, publication year, journal title, volume, issue, page numbers, DOI, and formatting, in accordance with the journal's reference style.